

MONCHO UNSCHOOLING

Dinosaurs

Unit Study

“Step into the shoes of a fossil hunter and discover that the ground under your feet holds the stories of giants who once shook the Earth.”

Biology | Earth Science | Astronomy | Math | Language Arts | Geography | Art & Creativity | History | Music | Movement & Body

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🧭 **How to use this guide:** This is not a workbook, and there's no answer key — on purpose! Every challenge is an invitation. Pick the ones that spark curiosity, skip the rest, and go find the answers by doing. **AI plans. Kids DO.** 🚀

🦴 BIOLOGY

Every living thing carries clues about where it came from, and some of those clues are hiding in your kitchen and in the ground.

🍗 Challenge 1: Bone Detective

🎯 Comparing Structures

Could a dinosaur's bones be hiding right inside your dinner plate? ❓

Birds are living dinosaurs. They descended from small feathered theropod dinosaurs, and their skeletons still show it. Cook a chicken wing or drumstick with an adult, let it cool, and clean the bones carefully. Wash your hands well afterward.

1. Lay the bones on a plate and find the long, thin bones and the hollow-looking tips.
2. Find a drawing of a theropod dinosaur skeleton in a library book or on a museum website.
3. Match each chicken bone to the same bone in the drawing: upper arm, forearm, thigh, shin.
4. Draw the chicken bones and the dinosaur skeleton side by side. Label every bone you matched.

Why: Birds kept many theropod bone shapes, but their bones became lighter and partly hollow, which helps flying animals stay light.

🌱 **Make it Easier:** Match just two bones, the thigh and the shin, and label only those.

🚀 **Level it Up:** Find a photo of a chicken's foot and count its toes. Count the toes on the dinosaur drawing too, then explain what the match tells you about how they are related.

🦷 Challenge 2: The Dinosaur Dentist

🎯 Form and Function

Why would a dinosaur need teeth shaped like a comb instead of a knife? ❓

You're the dinosaur dentist, and two patients need the perfect teeth. Scientists study coprolites, fossilized dinosaur poop, to learn what dinosaurs ate. Your job is to figure out which tools could do each eating job.

1. Gather a fork, scissors, a comb, and clothespins.
2. Gather foods: a leaf of lettuce, a carrot, a slice of cooked chicken or cheese, and a soft banana.
3. Test each tool on each food. Try stabbing, cutting, raking, and gripping.
4. Score each tool from 1 to 5 for every food in a test log.
5. Write your verdict: which tool is best for a plant-eater, and which for a meat-eater?

🌱 **Make it Easier:** Test only two tools and two foods, and circle the winner for each.

🚀 **Level it Up:** Invent a third dinosaur that eats both plants and meat, then pick the tool shapes that would make its ideal teeth.

🥚 Challenge 3: Guard the Nest

🎯 Animal Behavior

How do you keep a dozen eggs safe when hungry thieves are always nearby? ❓

Dinosaur eggs have been found in many places, including Mongolia's Gobi Desert, where Roy Chapman Andrews's expedition found nests in the 1920s. Now it's your turn to protect a nest.

1. Roll 6 to 8 clay eggs, each about the size of a walnut.
2. Build a nest from clay, soil, leaves, or sand. Arrange the eggs in a ring or spiral.
3. Decide who the egg thieves are, such as small mammals or lizards, and list two ways each might reach your nest.
4. Add protection to your nest, like a rim, a hidden spot, or leaves for camouflage.
5. Write a hatching-day plan: who guards, when you check the eggs, and what you do when the eggs crack.

🌱 **Make it Easier:** Build the nest with just 3 eggs and write a 3-step care plan.

🚀 **Level it Up:** Compare your plan with how a chicken or pigeon guards its eggs. Which ideas do the two plans share?

🪨 EARTH SCIENCE

Under your feet, hidden in layers of rock, are the stories of giants, and you're the fossil hunter who gets to read them.

🦴 Challenge 1: Crack Open Your Own Fossil

🎯 How fossils form

Why does a fossil look like the animal it came from, even though the animal is long gone? ❓

1. Roll a ball of modelling dough (or salt dough) and flatten it into a thick pancake.
2. Press a small shell or toy bone firmly into the middle, then lift it out carefully. The dent it leaves is your mold.
3. Mix a little plaster of Paris (or more salt dough) and spoon it into the mold. Tap the sides so it settles.
4. Let it harden. Plaster takes about 30 minutes, salt dough about a day.
5. Peel the dough away and look at your cast. Compare it with the real shell or bone. What details were copied?

Why: Real fossils form when a buried bone or shell is slowly replaced by minerals over millions of years, and most animals never become fossils at all. Your dough mold works like the mud around a bone, and the cast is the shape that stays behind.

🌱 **Make it Easier:** Skip the plaster and just press the shell into the dough. Draw what the mold looks like and what the real shell looked like.

🚀 **Level it Up:** Make three molds with three different objects, then ask a friend to guess each one from the cast alone, like a paleontologist.

🫙 Challenge 2: Dig Down Through Time

🎯 Rock layers

If you dig straight down, are you digging toward the past or the future?❓

1. Find a clear jar or glass. Gather sand, soil, rice and small gravel.
2. Pour in a thick layer of gravel. This is your oldest layer, so it goes in first.
3. Add a layer of sand, then press a small toy bone into the surface.
4. Cover the bone with a layer of rice, then top it all with a layer of soil.
5. Look through the glass. Which layer is oldest? Which is newest? Where would a fossil hunter have to dig to find your bone?

Why: Layers of sediment pile up one on top of the other over time, so deeper layers are usually older. The Mesozoic Era, the age of dinosaurs, lasted about 186 million years (about 252 to 66 million years ago). Its Triassic, Jurassic and Cretaceous periods are stacked in the rocks the same way.

🌱 **Make it Easier:** Use only three layers (sand, rice, soil) and label them oldest, middle and newest with sticky notes.

🚀 **Level it Up:** Add a second toy bone in a deeper layer. Then predict which of the two bones belongs to the older animal, and explain how you know.

🧶 Challenge 3: Build a Dino Time Rope

🎯 Time scale

Did Stegosaurus and T. rex ever meet, or did millions of years keep them apart? ❓

Unroll a long string or a toilet paper strip across a room or hallway. This is the Mesozoic Era, about 186 million years long.

1. Mark one end "start" and the other end "end of the dinosaurs."
2. Divide the rope into three colored sections for the Triassic, Jurassic and Cretaceous. Label each one.
3. Draw or stick a dinosaur in each section: small early dinosaurs in the Triassic, Stegosaurus at about 150 million years ago in the Late Jurassic, and T. rex at about 68-66 million years ago near the end.
4. Walk along your rope holding a toy dinosaur and stop at each one.
5. Write one sentence answering: Who lived closer to us, Stegosaurus or T. rex?
6. With an adult, search for a free "geologic time spiral" picture from the USGS (U.S. Geological Survey) and compare it with your rope.

🌱 **Make it Easier:** Use only two dinosaurs, Stegosaurus and T. rex, and a short rope with just a start and an end.

🚀 **Level it Up:** Add a tiny mark at the very end of your rope for "today" and measure the gap between T. rex and you. Then add one more animal of your choice and place it where it belongs.

🔭 ASTRONOMY

Space is not just far away: a giant rock from the sky changed the story of the dinosaurs forever.

🥄 Challenge 1: Crater Lab

🎯 Cause and Effect

Does a bigger space rock always make a bigger crater, or does something else matter too? ❓

1. Fill a baking tray with 2 cm of flour, smooth it flat, and dust the top with a thin layer of cocoa powder.
2. You're the impact scientist. Pick 3 balls of different sizes (a marble, a ping-pong ball, a golf ball). Predict which will make the biggest crater and sketch your guess.
3. Drop each ball from 30 cm. Lift it out carefully and take a photo or draw the crater, showing the cocoa "splash" around it.
4. Smooth the flour and repeat from 100 cm with the same ball. Does the same ball make a different crater?
5. Compare your predictions with your results.

Why: A faster, heavier rock carries more energy, so it digs a wider, deeper hole. Most scientists agree that an asteroid about 10 km wide struck Earth at Chicxulub, on Mexico's Yucatán Peninsula, about 66 million years ago, and non-bird dinosaurs went extinct around that time.

🌱 **Make it Easier:** Use just two balls and one drop height, and simply draw the two craters side by side.

🚀 **Level it Up:** Measure each crater's width with a ruler, then make a small chart of ball size, drop height and crater width to look for a pattern.

🧲 Challenge 2: Space Rock Detective

🎯 Observation and Testing

How could you tell a rock that fell from space apart from a plain old rock from your street? ❓

Meteorites are pieces of space rock that survived the fall through the sky and landed on Earth, and some are found by regular people! Gather 4-5 rocks from your yard or street, plus a refrigerator magnet.

1. Draw a checklist with three questions: Does the magnet stick? Is it heavy for its size? Does it have a dark, melted-looking crust?
2. Test each rock and tick the boxes. Hold two rocks of the same size, one in each hand, to compare how heavy they feel.
3. Look at free photos of real meteorites on the Natural History Museum (London) website or the NASA meteorite pages.
4. Compare your rocks with the photos and make a verdict for each one: space rock or Earth rock?

Why: Many meteorites contain iron, so they attract magnets and feel heavy, while most Earth rocks do not.

🌱 **Make it Easier:** Test only the magnet and the heaviness, and skip the crust.

🚀 **Level it Up:** Find the "Can I tell it's a meteorite?" tips on a museum website and add two more tests to your checklist, then explain why no single test is enough to be sure.

📐 MATH

Fossil hunters measure everything, and the numbers behind dinosaurs are huge, tiny, and surprising.

📏 Challenge 1: Giant Outline in the Yard

🎯 Measuring Length

How many of your own steps would it take to walk from the nose to the tail of one of the biggest animals that ever lived? ❓

Argentinosaurus, found in Argentina, is among the largest known dinosaurs, estimated at about 30-35 m long. Let's see what that looks like outside your door!

1. Pick a flat spot outside, like a yard, driveway, or quiet street. Ask an adult before you use chalk or tape.
2. Use a tape measure or a long string marked every meter. Mark a start line, then measure out 30 m and mark the end.
3. Draw or tape a straight line between the two marks. That is the dinosaur's length.
4. Walk the line heel-to-toe and count your steps. Then count how many bike lengths and how many family members lying head-to-toe it takes to match it.
5. Add a head at one end and a tail at the other, then give your giant a name.

🌱 **Make it Easier:** Use a 10 m piece of string instead and count how many of your steps fit. Then multiply by 3 to guess the full length.

🚀 **Level it Up:** Divide 30 m by your step length to predict the number of steps first. Then walk it and see how close your prediction was.

⚖️ Challenge 2: Raptors on the Scale

🎯 Division and Weight

How many dinosaurs the size of a turkey would it take to weigh as much as you? ❓

You're the raptor accountant, and your job is to count raptors by weight. Velociraptor, from Mongolia, was about the size of a turkey (roughly 2 m long and 15 kg), not the human-sized monster shown in films, and it had feathers.

1. Weigh a family member on a bathroom scale. Ask first, and write the number down.
2. Divide that weight by 15 to find how many Velociraptors it would take to match.
3. If there is a remainder, write it as "and a bit left over."
4. Try it with 3 more people or pets and make a small table of your results.

For example: 45 kg ÷ 15 kg = 3 raptors.

Why: Films made Velociraptor look huge, but fossils show it was only about as big as a turkey, so it takes many of them to equal one person.

🌱 **Make it Easier:** Use a number that divides evenly, like 30 kg, and count by 15s on your fingers (15, 30, 45...).

🚀 **Level it Up:** Compare yourself to two other dinosaurs at once, and find the difference in weight, not just the number of raptors.

👣 Challenge 3: Who Was Running?

🎯 Measuring and Speed

How can a line of old footprints tell you whether a dinosaur was strolling or sprinting? ❓

Dinosaur footprints (trackways) are real fossils, and scientists measure the distance between steps to estimate how fast the animal walked. The longer the stride, the faster the dinosaur!

1. Trace and cut out 6 footprint shapes from paper or cardboard, each about the size of your hand.
2. Lay them in a line on the floor with steps 30 cm apart. Label this track "Dino A."
3. Make a second track with steps 60 cm apart ("Dino B") and a third with steps 100 cm apart ("Dino C").
4. Measure each gap with a ruler or tape and write it down.
5. Rank the dinosaurs from slowest to fastest. Then walk beside each track and notice how your own steps stretch out as you go faster.

Why: When an animal speeds up, its legs swing farther with each step, so a longer gap between footprints points to a faster walker.

🌱 **Make it Easier:** Use only two tracks, one with short steps and one with long steps, and just decide which dinosaur was faster.

🚀 **Level it Up:** Add up the total distance covered by each track by multiplying the gap by the number of steps. Then invent a fourth dinosaur whose track is exactly twice as long as Dino A's.

📖 LANGUAGE ARTS

Dinosaur names, letters and dictionary pages are all ways of using words to bring giants back to life.

📖 Challenge 1: Name a New Dinosaur

🎯 Greek Word Roots

If every part of a dinosaur's name is a clue, what could a name tell you before you ever saw the animal? ❓

Triceratops means "three-horned face." Scientists build dinosaur names from Greek pieces, like tri- (three), cerat- (horn) and -ops (face).

1. Copy the three pieces above, then look up 3 more roots in a dictionary or book, such as mega- (big), saurus (lizard) and ptero- (wing).
2. Mix two or three roots to invent a name for a dinosaur you dreamed up.
3. Write what your name means in plain English.
4. On an index card, write the name large, add the meaning underneath and explain in one sentence why those roots fit your creature.

🌱 **Make it Easier:** Use only two roots and write the meaning as a short phrase, like "big lizard."

🚀 **Level it Up:** Invent a second dinosaur that is the "cousin" of the first, with a name that shares one root, and explain how the two names show they are related.

📖 Challenge 2: Letter From the Cliffs

🎯 Letter Writing

How would it feel to hold something nobody has seen in millions of years? ❓

Mary Anning was born in 1799 in Lyme Regis, England, and as a child she collected fossils from the seaside cliffs. Around 1811-1812 she and her brother found one of the first ichthyosaur skeletons known to science. Ichthyosaurs were marine reptiles, not true dinosaurs, but they lived in the same ancient world.

You are a young fossil collector in Lyme Regis, and you have just made a big discovery.

1. Jot down 5 senses words for the cliffs: how the wind sounds, how the rock feels, how the air smells.
2. Describe your find: what you saw first and what you did next.
3. Tell the museum how it felt to be the first person to see it in millions of years.
4. Write it as a neat letter on paper with a greeting, your message and a sign-off.

🌱 **Make it Easier:** Write only 5 sentences: what you found, where, and how you felt.

🚀 **Level it Up:** Write a reply from the museum, asking you three questions about your find, then answer them.

📖 Challenge 3: The Word "Dinosaur"

🎯 Word Origins

Why do you think a name invented almost two centuries ago is still the word we use today? ❓

The word "dinosaur" comes from Greek words meaning "terrible lizard." Richard Owen coined it in 1842.

1. Find the Greek roots deinos (terrible) and sauros (lizard), and write them in the corner of a sheet of paper.
2. Draw a line down the page. On the left, write the entry's parts: the word, how to say it (DY-nuh-sor), the part of speech and its meaning.
3. Add a short "word history" explaining who made it up, when and what it combines.
4. Add one sentence of your own using "dinosaur" and think about why the name stuck: it is short, dramatic and easy to remember.
5. Draw a small dinosaur in the margin.

🌱 **Make it Easier:** Fill in only the word, how to say it, the meaning and one example sentence.

🚀 **Level it Up:** Look up another word with the root -saur or deinos and add a "see also" line with its meaning.

🗺️ GEOGRAPHY

Fossil hunters travel the whole planet, because dinosaur bones are tucked away in rocks on almost every continent.

🌍 Challenge 1: Pin the Dig Sites

🎯 Map Reading

Why do dinosaur bones turn up in places as different as icy badlands and sandy deserts? ❓

You're the expedition planner for a team of fossil hunters, and you need a map of where to dig.

1. Draw or print a blank world map and label the continents.
2. Find four famous fossil sites in an atlas or on a globe: the Gobi Desert, Alberta, Patagonia and Montana. Mark each with a star.
3. With a grown-up, browse the online collections of the Natural History Museum in London and the American Museum of Natural History in New York. Both display famous dinosaur skeletons, so pick one dinosaur from each museum.
4. Draw a small picture of a dinosaur next to the site where you think it fits best, and add a compass rose and your own map legend.

🌱 **Make it Easier:** Mark only two sites and use a printed map that already has the continent names on it.

🚀 **Level it Up:** Add a dotted line showing which two sites are closest together, then use the map scale to estimate how many kilometers apart they are.

🧭 Challenge 2: Gobi Expedition Prep

🎯 Trip Planning

What would you need to survive for two weeks in one of the driest, windiest places on Earth?❓

Your team is leaving for the Gobi Desert in Mongolia, and the packing list is up to you.

1. Look up the Gobi's weather with a grown-up. How hot does it get in summer, and how cold in winter? Write both down.
2. Trace a travel route on a world map from your home to Mongolia. List each stop: plane, then truck or jeep, then walking on foot.
3. Write a supply list under three headings: Water and Food, Clothing, and Digging Tools. Include at least four items under each heading.
4. Circle the three items you could not live without, and be ready to explain your choices.

Why: Deserts swing between hot days and freezing nights because there are no clouds or plants to hold the heat in after sunset.

🌱 **Make it Easier:** Pick just one season, then write a list of only 8 items you would pack.

🚀 **Level it Up:** Add a 5-day schedule showing where you'd camp each night, and count how much water your team of four needs for the 5 days if each person drinks 4 liters a day.

🎨 ART & CREATIVITY

Dinosaurs left behind bones, footprints and mysteries, and artists turn those clues into creatures you can almost touch.

🦕 Challenge 1: Tiny-Headed Stegosaurus

🎯 Sculpting proportions

How small could a head be on a body as big as a bus? ❓

Time to model a plated giant with a surprisingly small head.

1. Mix 1 cup flour, ½ cup salt and about ½ cup water into salt dough, or grab air-dry clay.
2. Roll a big oval body about the size of your fist. Add four thick legs, with the back legs longer than the front ones.
3. Shape a long tail and push 3-4 spikes into the end of it.
4. Make the head tiny: no bigger than a walnut or a small grape. Add a pointy beak.
5. Press a row of flat triangle plates down the back, tallest in the middle.

Why: A real Stegosaurus had a brain about the size of a lime, while its body was as long as a bus. That is why your head must look comically small.

🌱 **Make it Easier:** Build only the body, head and four stubby legs, and skip the spikes. Stick on paper plates instead of sculpting them.

🚀 **Level it Up:** Make a second model with a head that is the "normal" size for a body like that. Set both side by side and describe how different they look.

🖍️ Challenge 2: Paleo-Artist at the River

🎯 Scientific illustration

What would a river look like when a giant swimming dinosaur lived in it? ❓

You're the paleo-artist hired to draw a Spinosaurus, a dinosaur that likely hunted around rivers. Fossil hunters in places like Dinosaur Provincial Park in Alberta, Canada, a UNESCO World Heritage Site with fossils from dozens of dinosaur species, rely on artists like you to show what their finds looked like alive.

1. Look up clues in a library book or a trusted kids' encyclopedia: Spinosaurus had a tall sail on its back, a long narrow snout and paddle-like feet.
2. Sketch your dinosaur in pencil, big enough to fill half a page.
3. Draw its river home: water, muddy banks, plants and a few fish.
4. Color it in, then label 3 parts that came from fossil clues (sail, snout, feet).
5. Show your picture to someone and explain why you drew each part the way you did.

🌱 **Make it Easier:** Trace a simple dinosaur outline, then draw the water and plants around it. Label only 1 part.

🚀 **Level it Up:** Add a second animal from the same river and explain which clue helped you decide what it ate.

🥔 Challenge 3: Footprint Stampede

🎯 Printmaking

How can one tiny stamp turn into a whole herd on the move? ❓

1. Cut a potato in half, or cut a kitchen sponge into a flat shape. Ask an adult to help with the knife.
2. Carve or trim a three-toed footprint into the flat face. A big rounded heel pad with three thick toes works well.
3. Dab the stamp in washable paint and press it on scrap paper to test it.
4. Unroll a long strip of paper, like the back of wrapping paper, and print a line of footprints all the way across.
5. Print a second, smaller set of prints beside the first, then add a third set going the opposite way. Space the steps closer together for small dinosaurs and farther apart for big ones.

🌱 **Make it Easier:** Skip carving. Paint your own foot or hand and stamp it instead, using a few toe marks.

🚀 **Level it Up:** Make two different stamps for two different species, and add tiny drawn plants or rocks between the tracks to build the landscape.

📜 HISTORY

Long before scientists studied dinosaurs, people had to dig, argue, and change their minds again and again, and history is the story of how we figured them out.

🎤 Challenge 1: The Dinosaur Memory Interview

🎯 Interviewing

How did one big discovery change what people could imagine about dinosaurs? ❓

In 1902, Barnum Brown found the first fossil Tyrannosaurus rex skeleton in Montana, USA. Almost 90 years later, in 1990, a skeleton nicknamed "Sue" was found in South Dakota, and today she stands in the Field Museum in Chicago. Now it's your turn to collect dinosaur history from someone close to you.

1. Pick a family member or friend and write 3 questions, like "What's your first dinosaur memory?" and "Which dinosaur was your favorite when you were little?"
2. Ask your questions and write down their answers in your own words.
3. Ask one extra question: "What did dinosaurs look like in the pictures you saw as a kid?"
4. Draw a line down the middle of a page. On the left, write what people pictured *before* big fossil discoveries. On the right, write what your interview showed you about dinosaurs today.

🌱 **Make it Easier:** Ask only 2 questions and draw your person's favorite dinosaur instead of writing the answers.

🚀 **Level it Up:** Interview two people from different generations, like a grandparent and a friend your age, and compare how their dinosaur pictures were different.

🪶 Challenge 2: Redesign the Dinosaur Exhibit

🎯 Comparing Evidence

Why would a scientist be happy to find out an old dinosaur drawing was wrong? ❓

You're the museum redesigner, and the old dinosaur poster in your hall needs an update. In 1861, workers in Germany found a raven-sized fossil in limestone near Solnhofen. It was Archaeopteryx, a bird-like dinosaur with feathers, and it helped scientists see that some dinosaurs looked very different from the scaly monsters in old pictures.

1. Look up an old dinosaur illustration in a library book, and notice the skin and the pose.
2. Look at a modern picture of a feathered dinosaur, and list 3 differences.
3. Fold a sheet of paper in half.
4. On the left, draw the dinosaur the old way. On the right, redraw it with feathers and an active pose.
5. Write a caption under each side that tells visitors what scientists believed then and now.

Why: new fossils with feather marks pressed into fine-grained rock showed scientists they had been missing part of the story, so they updated their pictures.

🌱 **Make it Easier:** Draw just one dinosaur and add feathers with colored pencils on a printed outline.

🚀 **Level it Up:** Add a third panel showing what you think scientists might discover next, and explain which clue from the fossils supports your idea.

🎵 MUSIC

Fossil hunters can't hear dinosaurs, but you can invent how they might have sounded by building instruments and rhythms inspired by their bodies.

🎺 Challenge 1: Build a Crest Trumpet

🎯 Pitch and Length

If a tube gets longer, what happens to the sound that comes out of it? ❓

Scientists think the hollow, curved crest on Parasaurolophus' head may have helped it make deep, booming calls. Time to test the idea with your own tube.

1. Grab a paper-towel tube and a paper cup. Ask an adult to help you cut a hole in the bottom of the cup, then tape the tube into it, so the cup is the "bell" of your trumpet.
2. Hum or say "hooooo" into the open end of the tube. Listen to the sound.
3. Cut the tube into two shorter lengths (ask an adult to help with scissors) and try each one. Which makes a higher sound? Which is lower?
4. Now compose a call using 3 sounds, like long-short-long. Draw it as dashes and dots, for example "— · —".
5. Record your call with a phone or tablet, then play it back and see if a friend can copy the pattern.

Why: sound comes from air vibrating inside the tube. A longer tube holds a longer column of air, which vibrates more slowly and makes a lower note.

🌱 **Make it Easier:** Skip the cutting. Use just one tube and change your voice from high to low, then draw a call with only 2 sounds.

🚀 **Level it Up:** Make 3 tubes of different lengths and order them from lowest to highest. Then compose a 6-sound call and write it out so another person can perform it from your notes.

🥁 Challenge 2: Giant Stomp vs. Fast Feet

🎯 Rhythm and Tempo

How can the same body make a giant sound slow and heavy, or small and quick? ❓

Brachiosaurus was a long-necked plant eater from the Late Jurassic. Picture a giant like that taking one slow, heavy step at a time. Now picture a Velociraptor sprinting past it.

1. Make a slow Brachiosaurus beat. Stomp one foot and count "1 ... 2 ... 3 ... 4" with a big pause between stomps.
2. Make a fast Velociraptor beat. Tap your fingers on your knees as quickly as you can in a steady pattern, like "tap-tap-tap-tap."
3. Add one more sound to each one, such as a clap for the Brachiosaurus and a snap for the Velociraptor.

4. Perform both together. Try doing the Brachiosaurus stomps with your feet while your hands play the Velociraptor, or ask a partner to play one part while you play the other.
5. Practice until you can play 8 beats of each without losing your place.

Why: tempo is how fast or slow the beat is. A steady slow beat feels heavy and large, while a fast beat feels light and small.

🌱 **Make it Easier:** Play only the Brachiosaurus stomp for 4 beats, then switch to the Velociraptor taps for 4 beats, one after the other.

🚀 **Level it Up:** Add a third dinosaur with its own tempo and sound, such as a medium-speed walker. Then perform all three together and write the rhythm as a pattern of big and small symbols.

🏃 MOVEMENT & BODY

Dinosaurs didn't just exist, they moved, and your own body can help you figure out how.

🦖 Challenge 1: Tiny Arms, Giant Balance

🎯 Balance and Coordination

If you had a huge heavy tail and tiny arms, how would you keep from tipping over?❓

Time to move like a T. rex!

1. Roll a bath towel tightly and tuck one end into the back of your waistband so it sticks out behind you like a heavy tail.
2. Curl your arms up close to your chest, with your hands small and bent, like a T. rex.
3. Lean your chest forward until the towel tail lifts up behind you. Hold the pose for 10 seconds without wobbling.
4. Now walk across the room in slow, heavy steps, keeping your head forward and your tail steady.
5. Drop the towel and run on your toes with your arms swinging, light and quick like a Velociraptor-style sprinter. Feel how different it is!
6. Put it together: create a short sequence of 4 to 6 moves that shows how each dinosaur balances and moves.

Why: A long tail works like the other end of a seesaw. It balances the weight of the head and chest out front, so the body can stay level over the hips.

🌱 **Make it Easier:** Skip the towel tail. Just curl your arms up, lean forward, and take 5 slow, heavy steps like a T. rex, then 5 quick light steps.

🚀 **Level it Up:** Try doing your sequence with the towel tail on while balancing on one foot for 5 seconds in the middle. Notice what happens when you swing the tail to one side, and teach your sequence to someone else.

🦕 Challenge 2: Dino Coach's Obstacle Course

🎯 Agility and Planning

What would your body have to do differently to move like a slow giant, a speedy runner, and a tall grazer? ❓

You're the dino coach, and today you're training a team of new dinosaurs.

1. Clear a safe space indoors or outside and mark 3 stations with cushions, shoes, or chalk.
2. Station 1, Slow Stomp: walk heavily with big, slow steps, lifting each knee high and pressing each foot down firmly.
3. Station 2, Quick Dash: crouch low, then sprint 10 steps to the next marker as fast as you safely can.
4. Station 3, Long-Neck Stretch: stand tall, reach both arms up high, stretch your neck forward and slowly sway side to side as if you were nibbling treetops.
5. Walk through the whole course in order, then run it again as your own made-up dinosaur.
6. Give a friend or family member a turn, and coach them through each station with a loud, clear voice!

🌱 **Make it Easier:** Use just two stations, a slow stomp and a quick dash, and do each one only once.

🚀 **Level it Up:** Add a fourth station for a new dinosaur move you invent, like a sneaky tiptoe, and time how long each run takes. Can you beat your own time without losing any of the stomps, dashes, or stretches?